

Toward Stable and Identifiable Bias Directions in Large Language Models

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1. Motivation

Recent work (Kim, Evans & Schein, ICLR 2025) demonstrates that high-level social constructs can emerge as linear directions in representation space of large language models (LLMs). These findings suggest that interpretable structure may exist within deep neural representations and can be manipulated via linear interventions.

However, a key open question remains:

Are such directions statistically stable and identifiable, or are they partially artifacts of prompt selection, sampling noise, or layer-specific effects?

This question is critical for both:

- interpretability (whether a direction reflects a true latent construct), and
- intervention (whether modifying that direction has consistent semantic effects).

2. Research Question

This project investigates:

1. Existence vs. Stability

- Do bias directions (e.g., gender, occupation stereotypes) consistently emerge across resampled datasets?

2. Layer-wise Identifiability

- Are certain Transformer layers more likely to encode stable and interpretable directions?

3. Robustness of Linear Structure

- To what extent does the assumption of a single linear direction hold under perturbations?

3. Methodology

3.1 Representation Extraction

- Extract hidden states from each Transformer layer for a controlled set of prompts
- Prompts vary only in sensitive attributes (e.g., gendered names) while holding semantic structure constant

3.2 Linear Probe Estimation

- Train linear classifiers (logistic regression) to identify candidate bias directions (θ)
- Normalize coefficients to obtain unit direction vectors

3.3 Stability Analysis (Core Contribution)

To assess identifiability, I introduce:

(a) Bootstrap Stability

- Resample training data multiple times
- Estimate $\theta^{(b)}$ for each bootstrap sample
- Measure pairwise cosine similarity: $[\text{Stability}] = \mathbb{E}_{i \neq j} [\cos(\theta^{(i)}, \theta^{(j)})]$

(b) Cross-layer Consistency

- Compare θ across layers
- Identify layers where representations are both:
 - separable (high probe accuracy)
 - stable (high similarity)

3.4 Intervention as Validation

- Apply inference-time intervention: $x_{\text{new}} = x + \alpha \sigma \theta$
- Evaluate whether stable directions correspond to consistent semantic shifts

4. Expected Contributions

This work reframes bias analysis in LLMs as a statistical estimation problem:

- Distinguishes between:
 - discoverable directions vs. reliable directions
- Introduces stability as a necessary criterion for interpretability
- Provides a diagnostic framework for evaluating linear representation assumptions

5. Broader Impact

Understanding whether bias directions are stable has implications for:

- Trustworthy interpretability
- Reliable bias mitigation
- General representation learning theory

If instability is observed, it may suggest that:

commonly used linear interventions operate on directions that are not uniquely identifiable.

6. Current Status

- Conceptual framework established
- Experimental pipeline under development (layer-wise probing + bootstrap analysis)
- Preliminary implementation completed for small-scale models

7. Future Directions

- Extend analysis to:
 - multi-attribute bias (race, age, intersectionality)
 - non-Transformer architectures (e.g., state space models)
- Formalize identifiability conditions for representation directions
- Connect stability to causal interpretability

8. Alignment with Your Work

This project directly builds upon your work on linear representations in LLMs, with a focus on:

moving from existence to statistical reliability and identifiability of representation directions.

I would be very interested in further developing this line of work under your guidance.

